

```

class Student {
// Public property - accessible everywhere
public name: string;
// Private property - only within Student class
private age: number;
// Protected property - within Student and child classes
protected address: string;
// Constructor to initialize properties
constructor(name: string, age: number, address: string) {
this.name = name;
this.age = age;
this.address = address;
}
// Public method to display student info
public displayInfo(): void {
console.log(`Name: ${this.name}`);
console.log(`Age: ${this.age}`);
console.log(`Address: ${this.address}`);
}
// Public method to get age (since age is private)
public getAge(): number {
return this.age;
}
// Public method to set age
public setAge(newAge: number): void {
if (newAge > 0 && newAge < 120) {
this.age = newAge;
}
}
}

// Create a student object
const student1 = new Student("Rahul Sharma", 20, "Mumbai, Maharashtra");
// Access public property directly
console.log(student1.name); // Works!
// Access private property directly
//console.log(student1.age); // ERROR! - age is private
// Access private property through public method
console.log(student1.getAge()); // Works!
// Access protected property directly
// console.log(student1.address); // ERROR! - address is protected
// Call public method
student1.displayInfo(); // Works!

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// Update age using setter
student1.setAge(21);
console.log(student1.getAge()); // Output: 21
```

```
class TE_Student extends Student {
// Own properties
public semV_Percentage: number;
public semVI_Percentage: number;
public branch: string;
private rollNumber: string;
// Constructor
constructor(
name: string,
age: number,
address: string,
semV: number,
semVI: number,
branch: string,
rollNumber: string
) {
// Call parent class constructor
super(name, age, address);
// Initialize own properties
this.semV_Percentage = semV;
this.semVI_Percentage = semVI;
this.branch = branch;
this.rollNumber = rollNumber;
}
// Method to calculate average percentage
public calculateAverage(): number {
return (this.semV_Percentage + this.semVI_Percentage) / 2;
}
// Method to check if student passed
public hasPassed(): boolean {
return this.semV_Percentage >= 40 && this.semVI_Percentage >= 40;
}
// Method to get grade based on average
public getGrade(): string {
const avg = this.calculateAverage();
if (avg >= 75) return "Distinction";
if (avg >= 60) return "First Class";
if (avg >= 50) return "Second Class";
if (avg >= 40) return "Pass Class";
return "Fail";
}
```

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}
// Getter for roll number
public getRollNumber(): string {
return this.rollNumber;
}
// Override parent method to display TE student info
public displayInfo(): void {
console.log("=== TE Student Information ===");
console.log(`Name: ${this.name}`);
console.log(`Age: ${this.getAge()}`);
// Can access protected 'address' from parent class!
console.log(`Address: ${this.address}`);
console.log(`Roll Number: ${this.rollNumber}`);
console.log(`Branch: ${this.branch}`);
console.log(`Sem V: ${this.semV_Percentage}%`);
console.log(`Sem VI: ${this.semVI_Percentage}%`);
console.log(`Average: ${this.calculateAverage()}%`);
console.log(`Grade: ${this.getGrade()}`);
console.log(`Status: ${this.hasPassed() ? "PASSED" : "FAILED"}`);
}
}
// Create TE_Student objects
const teStudent1 = new TE_Student(
"Aakanksha Bhope",
19,
"Pune, Maharashtra",
78.5, // Sem V percentage
82.0, // Sem VI percentage
"Information Technology",
"TE-IT-101"
);
const teStudent2 = new TE_Student(
"Arjun Kumar",
20,
"Nagpur, Maharashtra",
65.0,
70.5,
"Information Technology",
"TE-IT-045"
);
const teStudent3 = new TE_Student(
"Sneha Desai",
21,
"Nashik, Maharashtra",

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55.0,  
68.5,  
"Electronics Engineering",  
"TE-EXTC-067"  
);
```

```
class BE_Student extends Student {  
    // Own properties  
    public semVII_Percentage: number;  
    public semVIII_Percentage: number;  
    public branch: string;  
    private rollNumber: string;  
    // Constructor  
    constructor(  
        name: string,  
        age: number,  
        address: string,  
        semVII: number,  
        semVIII: number,  
        branch: string,  
        rollNumber: string  
    ) {  
        // Call parent class constructor  
        super(name, age, address);  
        // Initialize own properties  
        this.semVII_Percentage = semVII;  
        this.semVIII_Percentage = semVIII;  
        this.branch = branch;  
        this.rollNumber = rollNumber;  
    }  
    // Method to calculate average percentage  
    public calculateAverage(): number {  
        return (this.semVII_Percentage + this.semVIII_Percentage) / 2;  
    }  
    // Method to check if student passed  
    public hasPassed(): boolean {  
        return this.semVII_Percentage >= 40 && this.semVIII_Percentage >= 40;  
    }  
    // Method to get grade based on average  
    public getGrade(): string {  
        const avg = this.calculateAverage();  
        if (avg >= 75) return "Distinction";  
        if (avg >= 60) return "First Class";  
        if (avg >= 50) return "Second Class";  
    }  
}
```

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if (avg >= 40) return "Pass Class";
return "Fail";
}
// Getter for roll number
public getRollNumber(): string {
return this.rollNumber;
}
// Override parent method to display TE student info
public displayInfo(): void {
console.log("=== BE Student Information ===");
console.log(`Name: ${this.name}`);
console.log(`Age: ${this.getAge()}`);
// Can access protected 'address' from parent class!
console.log(`Address: ${this.address}`);
console.log(`Roll Number: ${this.rollNumber}`);
console.log(`Branch: ${this.branch}`);
console.log(`Sem VII: ${this.semVII_Percentage}%`);
console.log(`Sem VIII: ${this.semVIII_Percentage}%`);
console.log(`Average: ${this.calculateAverage()}%`);
console.log(`Grade: ${this.getGrade()}`);
console.log(`Status: ${this.hasPassed() ? "PASSED" : "FAILED"}`);
}
}
const beStudent1 = new BE_Student(
"Sejal Surve",
23,
"Nashik, Maharashtra",
55.0,
68.5,
"Information Technology",
"TE-IT-067"
);
// Display all student information
teStudent1.displayInfo();
console.log("\n");
teStudent2.displayInfo();
console.log("\n");
teStudent3.displayInfo();
console.log("\n")
beStudent1.displayInfo();

```

Rahul Sharma

20

Name: Rahul Sharma

Age: 20

Address: Mumbai, Maharashtra

21

=== TE Student Information ===

Name: Aakanksha Bhope

Age: 19

Address: Pune, Maharashtra

Roll Number: TE-IT-101

Branch: Information Technology

Sem V: 78.5%

Sem VI: 82%

Average: 80.25%

Grade: Distinction

Status: PASSED

=== TE Student Information ===

Name: Arjun Kumar

Age: 20

Address: Nagpur, Maharashtra

Roll Number: TE-IT-045

Branch: Information Technology

Sem V: 65%

Sem VI: 70.5%

Average: 67.75%

Grade: First Class

Status: PASSED

=== TE Student Information ===

Name: Sneha Desai

Age: 21

Address: Nashik, Maharashtra

Roll Number: TE-EXTC-067

Branch: Electronics Engineering

Sem V: 55%

Sem VI: 68.5%

Average: 61.75%

Grade: First Class

Status: PASSED

=== BE Student Information ===

Name: Sejal Surve

Age: 23

Address: Nashik, Maharashtra

Roll Number: TE-IT-067

Branch: Information Technology

Sem VII: 55%

Sem VIII: 68.5%

Average: 61.75%

Grade: First Class

Status: PASSED

=== Code Execution Successful ===